7.10.4 Pathogenesis

The pathogenic mechanism of *Vibrio cholerae* is discussed below in flowchart 7.7.

Source of Infection – contaminated water or food

Route of entry – fecal – oral route

Site of infection – small intestine

Incubation period – few hours to 5 days (usually 2–3 days)

Vibrio cholerae causes cholera, which is an acute diarrheal disease.

↓

In humans, *Vibrio* enter orally through contaminated water or food. The ingested pathogens pass through the acid barrier of the stomach & multiply in the small intestine.

↓

In the small intestine, the *Vibrio* penetrate the mucous barrier & adhere to the microvilli of the epithelial cells & multiply there.

↓

The *Vibrio*is strictly epipathogen& do not penetrate deep into the guts and there is no bacteremia. The virulence of *Vibrio cholerae* is due to cholera toxin (the mechanism is described earlier in the section 7.10.3 – enterotoxinmode of action)

↓

The toxin also inhibits intestinal absorption of sodium and chloride.

↓

The clinical manifestations & complications are due to massive water and electrolyte depletion.

↓

The voided fluids areodourless and contains flecks of mucus, hence it is known as Rice watery stool

Flowchart 7.7: Pathogenesis of *Vibrio cholerae*



The loss of water during cholera is about 20–30 litres per day.

HOTS

Why is cholera the most severe form of gastroenteritis?

7.10.5 Clinical Feature

Dehydration, anuria (absence of urine excretion), muscle cramps, hypokalemia (low blood potassium) & metabolic acidosis (low serum concentration of bicarbonates).

7.10.6 Laboratory Diagnosis

Specimen: Stool

Direct microscopy: It is not a reliable method for rapid diagnosis, the characteristic darting motility of the vibrio can be observed under dark – field microscope.

Culture: Stool sample is directly inoculated on MacConkey agar and TCBS agar. The plates are examined after overnight incubation at 37°C for typical colonies of *Vibrio cholera*, and the colonies are identified by gram staining and oxidasetest.

7.10.7 Prophylaxis

1. **General Measures**
   - Purification of water supplies
   - Improvement of environment sanitation
   - Infected patients should be isolated, and their excreta must be disinfected

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